Contact

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www.linkedin.com/in/peterbuilds (LinkedIn)

Top Skills

Innovation Management

Director level

Educational Consulting

Languages

Spanish (Native or Bilingual)

Italian (Limited Working)

English (Native or Bilingual)

Certifications

Programming and the Web for Beginners

Java Programming: Solving Problems with Software

Programming for Everybody (Getting Started with Python)

Peter Perez

Director of Innovation @ Belen Jesuit Prep | Leading Strategy, Curriculum & Operations for a K-12 Innovation Center | FRC Robotics Coach | Fab Academy Instructor | AP Physics C & Engineering Faculty

Hialeah, Florida, United States

Summary

With over 7 years of experience at Belen Jesuit Preparatory School, I currently serve as the Director of Innovation, leading strategy, curriculum, and operations for the Roberto C. Goizueta Innovation Center. My role includes setting the vision for our K-12 fabrication and innovation hub, managing equipment and safety standards, and overseeing a team to scale innovation programming across the school. I also coach the FRC Team 10177 Cyberclaws and teach AP Physics C and engineering coursework, integrating maker pedagogy into core STEAM instruction. As a Fab Lab Instructor at Moonlighter FabLab for over 3 years, I design and deliver hands-on digital fabrication workshops and camps, facilitating access to maker education for diverse learners. I also contribute to curriculum development, educator training, and program strategy as a member of the Advisory Board. Skilled in innovation management, educational consulting, and digital fabrication, I am passionate about fostering technical confidence and supporting creative problem-solving through project-driven learning experiences.

Experience

Belen Jesuit Preparatory School

7 years 4 months

Director of Innovation

June 2026 - Present (6 months)

Miami, Florida, United States

Lead strategy, curriculum, and operations for the Roberto C. Goizueta Innovation Center, a K-12 fabrication and innovation hub serving the full Belen student body.

Set vision and oversee operations for the Innovation Center, including equipment, space planning, and safety standards

Lead a team structure across operations and curriculum to scale Innovation Center programming school-wide
Own the Innovation Cohort, an enrichment program developing advanced student fabrication and design skills
Coach FRC Team 10177 Cyberclaws, rebuilding the program's rookie foundation toward competitive readiness
Teach AP Physics C and engineering/fabrication coursework, integrating maker pedagogy into core STEAM instruction
Pursue grant funding and represent the Innovation Center in school administration and Dean of Faculty planning

Associate STEAM Director / Fablab Specialist

November 2023 - June 2026 (2 years 8 months)
Miami, Florida, United States

Scaled a two 3D printer maker program into a fully equipped 9000 sq.ft K-12 fabrication lab, integrating CNC machining (ShopBot), laser cutting, 3D printing, embedded systems, and CAD, creating an accessible, standards-aligned STEAM environment for all students.

Designed and implemented a comprehensive STEAM curriculum aligned with NGSS and CTE standards, embedding project-based learning and design thinking frameworks across grades 6-12 to connect classroom work to real-world engineering practice.

Serve as STEAM liaison on the school-wide Math Curriculum Committee, contributing to a full restructuring effort designed to support every level of learner while expanding the academic ceiling of the program to include multivariable calculus and differential equations, reflecting a commitment to both equity of access and rigor of opportunity.

Led professional development for 50+ faculty members in digital fabrication and design thinking methodologies, contributing to a 25% increase in student engagement across STEAM subjects and school-wide adoption of experiential learning practices.

Directed FRC Team 10177 Cyberclaws and multiple FTC robotics teams, guiding students through competitive engineering design cycles and pursuing NASA Robotics Alliance Project grant support for program sustainability and growth.

Developed and launched updated course offerings in robotics programming (XRP kits), introductory coding, and 3D modeling, keeping curriculum current with emerging tools and aligned with workforce readiness expectations.

Established community partnerships with local industry and maker organizations (including Moonlighter Makerspace) to expand student access to real-world fabrication experiences and mentorship beyond the school building.

Presented at the Faith Forward national conference ("Making a Makerspace: From Cart to Center"), sharing a replicable model for growing vision-driven, resource-conscious makerspaces in K-12 settings.

Assistant STEAM Director

August 2019 - December 2023 (4 years 5 months)

Miami, Florida

Curriculum Leadership: Develop and implement interdisciplinary STEM/ STEAM curricula that integrate digital fabrication, product design, and robotics for grades 6-12. Incorporate design thinking and client-centered approaches into classroom projects, enhancing real-world problem-solving skills.

Program & Project Management: Lead end-to-end project management of student-centered initiatives, including robotics competitions and maker challenges. Oversee the planning, execution, and evaluation of school-wide STEAM events and workshops.

Technical Innovation: Manage the school’s maker space and fabrication lab, ensuring students and faculty have access to 3D printers, laser cutters, and CAD software (Onshape, SolidWorks). Guide students in product design and rapid prototyping, from ideation to functional prototypes.

Professional Development & Collaboration: Conduct workshops to train faculty on integrating technology and design thinking into their curricula. Collaborate with external partners to bring real-world STEM/STEAM applications into the classroom.

Community Engagement: Develop partnerships with local industry leaders to create mentorship opportunities and exposure to emerging technologies. Support student teams in robotics competitions, including FIRST Robotics, promoting leadership and teamwork.

Key Achievements:

Enhanced Student Engagement: Increased participation in STEAM programs by 40% through innovative curriculum redesign and hands-on learning opportunities.

Facility Improvements: Led the upgrade of the school’s maker space, integrating state-of-the-art equipment for digital fabrication and product development.

Faculty Empowerment: Facilitated professional development sessions attended by over 200 educators, resulting in school-wide adoption of design thinking methodologies.

Community Partnerships: Established local partnerships, enhancing students' exposure to real-world engineering and design challenges.

Moonlighter FabLab

Fab Lab Instructor

June 2023 - Present (3 years 6 months)

Miami Beach, Florida, United States

Develop and deliver hands-on FabLab curriculum and camp instruction for learners of all ages, designing experiences that build technical confidence through iterative, project-driven making.

Facilitate workshops in digital fabrication tools (3D printing, laser cutting, CNC, electronics) for diverse community audiences, expanding access to maker education beyond traditional school environments.

Author grant proposals to sustain and grow Moonlighter's educational programs, connecting community-based maker education to funding streams in workforce development, STEAM equity, and informal learning.

Serve on the Advisory Board, contributing strategic guidance on curriculum design, educator training, and program development to one of Miami's leading community fabrication labs.

Provide technical mentorship to students and interns, modeling the iterative design process and building the next generation of maker-educators in South Florida's growing makerspace ecosystem.

Florida International University - College of Engineering & Computing
Graduate Research Assistant/Expert
January 2021 - June 2023 (2 years 6 months)
Miami, Florida, United States

Graduate Research Assistant working under the CELL-MET project, a National Science Foundation (NSF) Engineering Research Center (ERC).

Currently assigned under the Workforce Development thrust focusing on K12, Undergraduate, Graduate, and Postdoctoral development in Cell and Tissue Engineering Concepts.

IMMACULATA LA SALLE HIGH SCHOOL INC
Engineering STEAM Focus Coordinator
June 2016 - August 2019 (3 years 3 months)
Miami, Florida

In my role, I direct the school's Engineering STEAM Program, which includes curriculum development and execution of four STEAM Engineering Courses (Introduction to Engineering; Robotics/Electronics; Engineering Principles; and Engineering Design and Development). I manage the design and maintenance of the STEM Lab/Maker Space, ensuring inventory of needed parts and supplies, and providing user training. I also coordinate STEAM committee meetings for program planning and collaborate with the STEAM program lead and Dean of Technology to assure program functionality to accreditation standards. I proudly serve as mentor to my students, providing guidance and direction for long term projects.I also:
• Operate extensively and instruct students in the use of 3D printing technology (Lulzbot, Formlabs Form2).
• Manage progress records to ensure college acceptance of 80 students.
• Serve as Lead Mentor for Bosco's Bots First Tech Challenge Team-7029, National Technical Honor Society Lead Mentor, and Entrepreneurship Club Coach.
My achievements in this role include:
• Effectively remodeled advanced STEM learning space for less than $13k.
• Increased size of Engineering program by 150%.

Immaculata-La Salle
Science Teacher
August 2014 - July 2016 (2 years)
Miami, Fl

When I first began teaching at this high school, I planned, created, and delivered in-depth science lessons, study materials, quizzes, and examinations for students. I also thoroughly investigated my students' individual comprehension levels, examining course work deficiencies and behavior issues to determine if student required intervention. I served as representative and speaker for the Science Department, STEAM Committee Co-Moderator, and coach for the FIRST Tech Challenge Team - 7029 Bosco's Bots. In addition, I:

• Developed and edited Unit Maps for Biology and Physical Science.

• Mentored Robotics team to compete in state competitions three consecutive years.

University of Miami, Miller School of Medicine

Research Intern

May 2015 - June 2015 (2 months)

Miami-Fort Lauderdale Area

-Miami Project to Cure Paralysis Summer Internship

-Care and treatment of HEK293 and Schwann Cell culture.

Independent Living Systems, LLC

Call Center Agent

November 2013 - July 2014 (9 months)

Doral, Florida

I provided vital customer service by assisting members with meal benefit and order information via telephone for this role. I assisted the company in evaluating customer satisfaction by completing courtesy calls, satisfaction surveys, post discharge referrals, and nutritional assessments and managed the delivery drivers' end of day reports, rescheduling and cancelling orders, and verifying member refusals as needed. I received recognition for:

• Training new incoming agents in order management, customer service, and company policies and procedures.

• Maintaining highest productivity in call center over a two-week period.

University of Miami

Research Assistant

October 2010 - May 2012 (1 year 8 months)

Veterans Association Research Laboratories, Miami, Fl

In this position, I provided vital investigation assistance for stem cell research by expanding, treating, and managing care for mesenchymal and periodontal ligament stem cells. I applied mechanical and electrical bioreactors to the

treatment of cells and used an Arduino microprocessor in series with analog circuits to make a customizable output function. I made sure to preserve the research data by keeping detailed lab notebooks on concept design and research. My achievements in this role include:

• Developed an electrical bioreactor for use in the laboratory.

• Created a "six-well plate electrode" to stimulate six cell cultures simultaneously

Leading and Learning

Tutor

July 2010 - December 2010 (6 months)

Coral Gables, Florida

Tutoring assigned students in general and college math and science as well as ACT and SAT prep.

University of Miami, School of Architecture

Student Assistant

September 2009 - January 2010 (5 months)

Coral Gables, Florida

• Answered phone calls directed to the front desk of the School of Architecture.

• Sorted all mail for the professors.

• Performed clerical duties as assigned.

University of Miami General Biology Howard Hughes Medical Institute Lab

Student

August 2009 - December 2009 (5 months)

Coral Gables, Fl

Used Gel Electrophoresis to quantify Zebrafish proteins for a DNA sequencing lab

Learned the techniques to set up a Polymerase Chain Reaction for DNA sequencing

Fertilized Sea Urchin Eggs in vitro using frozen or freshly extracted gamete

Tested the effect of Microwave radiation and water column depth on the cell fertilization

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Education

University of Central Florida

Master of Science (MS), Industrial Engineering · (2015 - 2020)

Fab Academy

Fab Academy - 2024 Cohort, Digital Fabrication · (January 2024 - August 2024)

University of Miami

Bachelor of Science (BS), Biomedical/Medical Engineering · (2008 - 2012)